Light

GCSE Computer Science – Data Representation: Numbers (Set 2)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

Which number system do humans use in everyday life?

- **A.** Denary (base 10)
- **B.** Octal (base 8)
- **C.** Binary (base 2)
- **D.** Hexadecimal (base 16)

Q2 (1 mark)

In the binary number 1101, what is the place value of the leftmost 1?

- **A.** 1

- B. 2
- C. 8
- D. 4

Q3 (1 mark)

What is the denary value of the binary number 1111?

- A. 15
- B. 14
- C. 12
- D. 16

Q4 (1 mark)

What is the binary representation of the denary number 9?

- A. 1011
- B. 1010
- C. 1001
- D. 1100

Q5 (1 mark)

In hexadecimal, what denary value does the letter A represent?

- A. 10
- B. 6
- C. 1
- D. 11

Q6 (1 mark)

In hexadecimal, what denary value does the letter E represent?

- A. 5
- B. 11
- C. 15
- D. 14

Q7 (1 mark)

How many bits are in one byte?

- A. 2
- B. 8
- C. 4
- D. 16

Q8 (1 mark)

How can you tell if a binary number represents an odd or even denary value?

- ○ **A.** Check if the rightmost bit (LSB) is 0 or 1
- ○ **B.** Check if the leftmost bit is 0 or 1
- ○ **C.** Count the total number of 1s in the binary number
- ○ **D.** Add up all the place values

Q9 (1 mark)

What is the minimum denary value that can be stored in 8 bits using unsigned binary?

- ○ **A.** -128
- ○ **B.** -1
- ○ **C.** 1
- ○ **D.** 0

Q10 (1 mark)

How many binary bits does one hexadecimal digit represent?

- ○ **A.** 2
- ○ **B.** 8
- ○ **C.** 4

- ○ **D.** 16

Q11 (1 mark)

In binary addition, what is 1 + 1?

- ○ **A.** 0
- ○ **B.** 10 (write 0, carry 1)
- ○ **C.** 2
- ○ **D.** 1

Q12 (1 mark)

What is the result of the binary addition 1001 + 0100?

- ○ **A.** 1100
- ○ **B.** 1110
- ○ **C.** 1101
- ○ **D.** 1111

Q13 (2 marks)

Convert the denary number 201 to an 8-bit binary number.

- ○ **A.** 11001001

- B. 11001011
- C. 11010001
- D. 11010011

Q14 (2 marks)

Convert the binary number 10101010 to denary.

- A. 168
- B. 174
- C. 172
- D. 170

Q15 (2 marks)

Convert the hexadecimal number A7 to denary.

- A. 167
- B. 161
- C. 157
- D. 177

Q16 (2 marks)

Convert the denary number 178 to hexadecimal.

- A. B0
- B. B2
- C. B8
- D. BE

Q17 (2 marks)

Convert the binary number 11110001 to hexadecimal.

- A. E1
- B. F0
- C. FE
- D. F1

Q18 (2 marks)

Convert the hexadecimal number 9C to binary.

- A. 10011100
- B. 10011010
- C. 10011110
- D. 10011111

Q19 (2 marks)

What is the result of the binary addition 10110011 + 00101100?

- ○ **A.** 11011101
- ○ **B.** 11011110
- ○ **C.** 11100000
- ○ **D.** 11011111

Q20 (2 marks)

An 8-bit register performs the addition 11110000 + 00100000. Which statement is correct?

- ○ **A.** The result is 100010000 with overflow
- ○ **B.** The result is 100010000 without overflow
- ○ **C.** The result is 00010000 with overflow
- ○ **D.** The result is 00010000 without overflow

Q21 (2 marks)

What is the effect of a left binary shift by two places on the number 00000101?

- ○ **A.** The value is multiplied by 2 to give 00001010
- ○ **B.** The value is divided by 4 to give 00000001
- ○ **C.** The value is divided by 2 to give 00000010

- ○ **D.** The value is multiplied by 4 to give 00010100

Q22 (2 marks)

What is the effect of a right binary shift by one place on the number 00101000?

- ○ **A.** The value is doubled to 01010000
- ○ **B.** The value is halved to 00010100
- ○ **C.** The value is quartered to 00001010
- ○ **D.** The value remains unchanged at 00101000

Q23 (2 marks)

A file is 2.5 megabytes (MB) in size. How many kilobytes (KB) is this?

- ○ **A.** 250 KB
- ○ **B.** 25,000 KB
- ○ **C.** 2,560 KB
- ○ **D.** 2,500 KB

Q24 (3 marks)

A computer adds three 8-bit unsigned binary numbers: 01100100 + 01100100 + 01100100.
Analyse the calculation and determine if overflow occurs.

- ○ **A.** Result is 100101100; no overflow because the result is within 8 bits
- ○ **B.** Result is 11111111; no overflow because the result equals 255
- ○ **C.** Result is 00101100; overflow occurs because adding three copies of 100 exceeds 255
- ○ **D.** Result is 10010110; overflow occurs on the first addition only

Q25 (3 marks)

A programmer divides the binary number 11111111 by 4 using right shifts. Analyse the result and explain any loss of precision.

- ○ **A.** Result is 00111111 (63); no precision loss because 255/4 = 63.75 rounds to 64
- ○ **B.** Result is 00111111 (63); precision loss occurs because the fractional part 0.75 is truncated, and two 1-bits are shifted off
- ○ **C.** Result is 01000000 (64); no precision loss because the shift automatically rounds up
- ○ **D.** Result is 00111111 (63); no precision loss because all bits are preserved

Q26 (3 marks)

A computer's memory addresses are displayed in hexadecimal. The address range for a small memory block is 00 to 7F.
Analyse how many unique memory locations this represents and explain why hex is used.

- ○ **A.** 79 locations; hex is used because it shows the binary patterns directly

- ○ **B.** 127 locations; hex is used because addresses must start at 01
- ○ **C.** 128 locations (0-127); hex is used because it compactly represents 8-bit addresses in just 2 characters while converting easily to binary
- ○ **D.** 256 locations; hex is used because the full address space is always shown

Q27 (3 marks)

The binary pattern 10000001 can be interpreted in two ways: as an unsigned integer or using two's complement for signed integers. Analyse what denary value each interpretation gives.

- ○ **A.** Unsigned: 129; Signed: -129
- ○ **B.** Unsigned: 129; Signed: -127
- ○ **C.** Unsigned: 127; Signed: -128
- ○ **D.** Unsigned: 128; Signed: -1

Q28 (3 marks)

Convert the denary number 219 to binary and hexadecimal.
Verify your answers by converting the hexadecimal back to denary.

- ○ **A.** Binary: 11011101, Hex: DD; Verification: (13 x 16) + 13 = 208 + 13 = 221
- ○ **B.** Binary: 11011011, Hex: D11; Verification: (13 x 16) + 11 = 219
- ○ **C.** Binary: 11011011, Hex: DB; Verification: (13 x 16) + 11 = 208 + 11 = 219

○ **D.** Binary: 11010111, Hex: D7; Verification: (13 x 16) + 7 = 208 + 7 = 215

Q29 (3 marks)

A programmer wants to multiply a number by 6 using only binary shifts and addition.
Analyse how this can be achieved efficiently.

○ **A.** Shift left 2 times to get 4x, then shift the result left 1 time to get 8x, giving 8x total

○ **B.** Shift left 2 times to get 4x the original, then add the original shifted left once (2x), giving 4x + 2x = 6x

○ **C.** Shift left 3 times to get 8x, then subtract the original twice, giving 8x - 2x = 6x

○ **D.** It is impossible to multiply by 6 using only shifts and addition

Q30 (3 marks)

A smartphone app is advertised as 45 MB. A user's phone shows 47,185,920 bytes of storage used after installation.
Analyse whether the app matches its advertised size and explain any discrepancy.

○ **A.** The app matches the advertisement; the difference is due to using 1 MB = 1,000,000 bytes (decimal) versus 1 MB = 1,048,576 bytes (binary), giving 45 MB in binary terms

○ **B.** The app is smaller than advertised; 47,185,920 bytes = only 37 MB

○ **C.** The app is larger than advertised; 47,185,920 bytes = 47.2 MB, so the advertisement is false

○ **D.** The app is exactly 45 MB; the extra bytes are temporary installation files

Total: **55** marks • Time: **30 min**

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